

Apoorv Mahajan

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SUMMARY

UX Researcher and UX Engineer with 6+ years of experience specializing in human-computer interaction (HCI) for AR/VR, AI, and automotive systems. Expert in conducting mixed-methods research to define interaction paradigms for multi-modal (voice/visual) interfaces. Proven track record of translating complex user behavioral data into high-fidelity prototypes and production-ready systems that prioritize safety, accessibility, and cognitive load management.

EDUCATION

2023 - 2025 Master of Science (Human-Computer Interaction) at **DePaul University** (GPA: 3.9/4.0)

2019 - 2023 Bachelor's Degree at **Symbiosis Center for Media and Communication** (GPA: 3.6/4.0)

WORK EXPERIENCE

UX Engineer | Woven by Toyota | Ann Arbor, MI

November 2025 - Present

- Engineered a custom Python backend bridge integrating the CARLA autonomous driving simulator with an iPad-based HMI web application (React, TypeScript, Bootstrap), achieving a 1-second visual glance-to-comprehension rate outperforming the 2-second automotive industry standard.
- Directed Wizard of Oz validation studies by triggering context-aware HMI screens and voice clips to test driver reactions to unexpected edge cases; proved that AI coaching increased driver confidence, improving overall driving efficiency by 33% and skill acquisition by 20% in initial sessions (with a 5% compounding improvement thereafter).
- Evaluated multimodal feedback loops (voice-first and UI-based) against complex driving scenarios, optimizing speech parameters and interaction timing to successfully reduce measured driver cognitive load by 43%.
- Leveraged AI coding tools to rapidly script, install, and validate custom behavioral features and traffic environments in CARLA, significantly accelerating the deployment of high-fidelity human-interaction research scenarios.
- Architected a scalable design system utilizing design tokens and visual regression testing to guarantee strict accessibility compliance, streamlining the Figma-to-CSS handoff process for the engineering team.
- Collaborating on a critical GAIA funded project with multiple Toyota companies (TRI, TMNA and TCNA).
- Daily Tools: Figma, VSCode, Claude Code, Python, PyGame, CARLA Simulator, GitHub, HTML, CSS, JS, DoveTail.

UI/UX Designer | Omnia | Chicago, IL

Sep 2024 - November 2025

- **Spatial Computing & XR Infrastructure:** Engineered high-performance 3D spatial interfaces for Apple Vision Pro and Meta Quest utilizing Unity (uGUI, TextMeshPro) and VisionOS SDKs, ensuring low-latency interaction and high visual fidelity across iOS, macOS, and XR environments.
- **Multimodal AI Validation:** Developed high-fidelity experimental prototypes in SwiftUI and React to A/B test AI-powered audible and visual feedback loops, successfully reducing user error by 13% during complex task execution in high-stakes automotive settings.
- **AR Usability & Product Strategy:** Directed formative usability testing for spatial interaction models and AR control schemes, refining visual hierarchies to reduce customer confusion by 13% and drive a 25% increase in subscriber retention.
- **Automotive UI/UX Engineering (ChiPark & Carvo):** Designed and deployed real-time geolocation applications for CarPlay and mobile ecosystems; integrated government-sourced APIs to reduce parking search times by 15% and increase payment speeds by 30%, while concurrently driving a 15% increase in fleet dispatch monitoring efficiency.
- **Safety-Critical Simulation:** Implemented Apple HIG, WCAG 2.0, and DMV requirements across all automotive UI components, utilizing Unity-based simulation prototypes to validate driver-assist motion, responsiveness, and distraction mitigation.
- **Spatial Training Prototyping:** Architected "How2To," an AI-powered VisionOS platform delivering real-time, step-by-step spatial tutorials, leading cross-functional engineering sprints to translate complex Figma prototypes into

production-ready Unity assets.

- **VisionOS Fleet Management:** Prototyped advanced Vision Pro interfaces for spatial navigation, fleet management, and 3D object filtering to streamline swift asset tracking.

UX Researcher and UI Designer | Patched Reality | Chicago, IL

Aug 2024 - Sep 2024

- Reach Shoulder Health: Rebuilt Unity-driven VR healthcare interfaces (uGUI) for physiotherapy simulations, improving usability and interaction accuracy by 27 percent .
- Increased user retention among the elderly by 50 percent by introducing a level progression map
- Collaborated with developers to debug and profile Unity scene performance, ensuring smooth user interactions across hardware tiers
- Increased average physiotherapy session time by 20 percent by reducing UI related confusion
- Partnered with Unity engineers to align interface logic, user input handling, and rendering performance with UX goals
- SynCare: Conducted longitudinal user studies for AI-integrated EHR systems, identifying key friction points in medical data visualization that led to a 25 percent reduction in measured cognitive load.
- Concluded that motivational messaging to follow a prescribed diet increased healthy habits by 35 percent
- Reduced overall cognitive load caused by current EHR, medical, and pharmaceutical app flows by 25 percent

Founder and UX Consultant | Crazy Minds | Remote

Jul 2019 - Sep 2023

- Delivered 25+ UX/UI projects across fintech, health-tech, and e-commerce with measurable improvements
- Created modular design systems supporting developers' onboarding and faster product cycles
- Led user research and iterative redesign for mobile-first and web-first platforms
- Delivered Unity-based UI prototypes for AR/VR and SaaS clients, bridging front-end design with Unity interaction scripting for rapid iteration

LEADERSHIP AND COLLABORATION

- Planned, designed, collaborated and iterated on 3 B2B and 2 B2C products end-to-end with engineers to drive successful market launches
- Facilitated workshops and built trust with key stakeholders to align on product vision and implementation
- Developed and governed organization-wide Figma design systems (100+ reusable components), enforcing industry standards of consistency and reducing handoff/QA issues by over 25 percent
- Conducted development sprints using Jira and Confluence with developers for efficient usage of JS, HTML, CSS, React.js, Angular, Vue and SwiftUI to meet the design direction
- Presented design solutions to C-suite and senior leaders, communicating rationale and the impact of UI improvements
- Mentored visual design and UX Research interns

SKILLS

Research Methods: Quantitative & Qualitative Analysis, Longitudinal Studies, Contextual Inquiry, Eye-tracking Analysis

HCI Frameworks: Human Factors for Automotive, Spatial UI Heuristics, Cognitive Load Theory, WCAG 2.1/3.0.

Tools: Figma, SwiftUI, Storybook, Jira, Confluence

Development Collaboration: HTML, CSS, React.js, SwiftUI, UI Toolkit

Design Systems: WCAG, HIG, Modular Systems